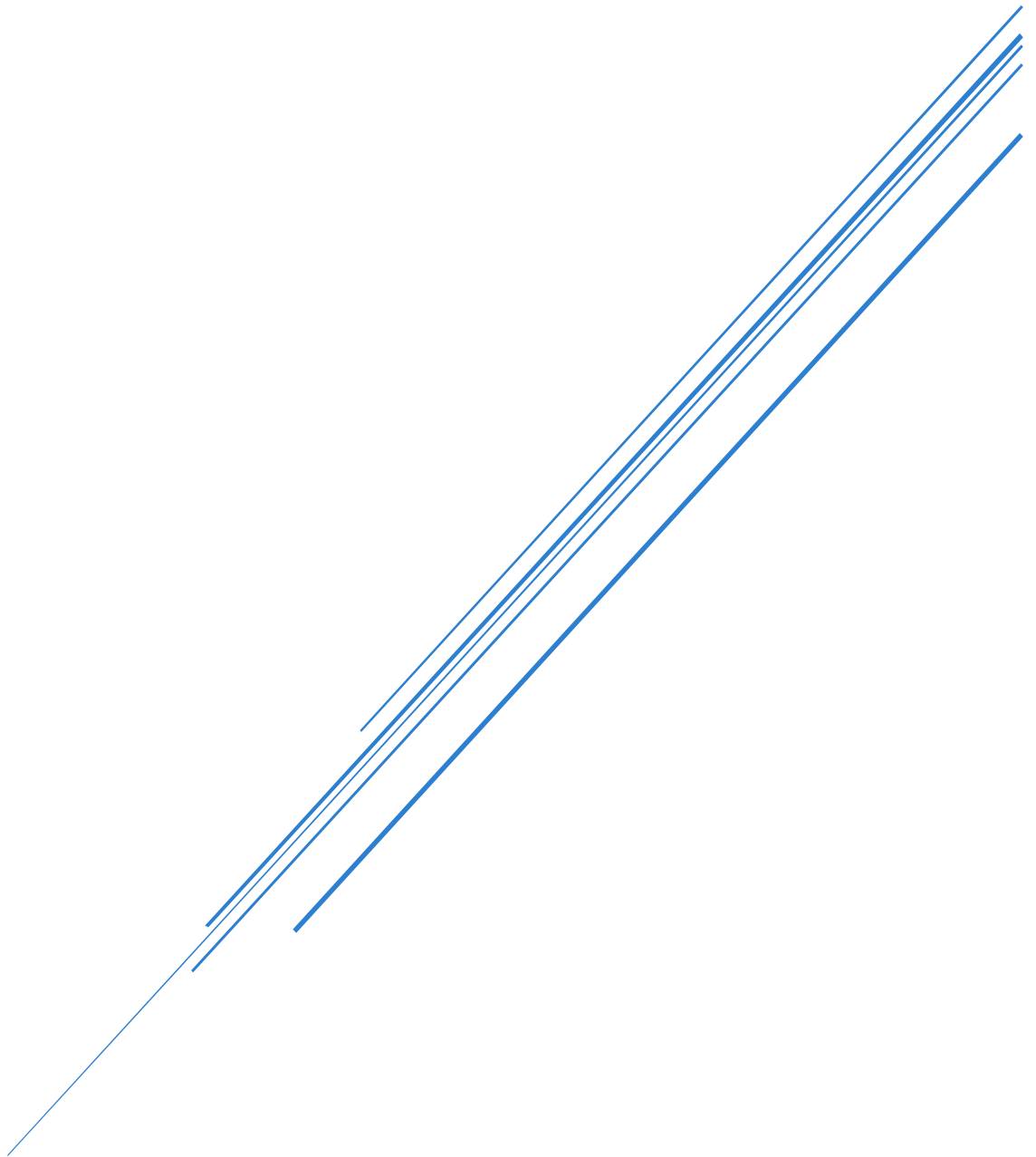


NHS ENGLAND

Capacity vs. Utilisation Study



Julita Drwiega

Context

The NHS is currently experiencing significant pressure as the number of GPs is decreasing, the number of patients per GP is increasing, and patient satisfaction is decreasing as a result (BMA). This leads to ambiguity regarding the effective use of network capacity and whether additional resources are genuinely required. The 5 Whys framework (see Appendix 1) was employed to identify the root causes of the issues and to establish project objectives.

This study aimed to analyse trends in appointment data to determine if capacity limits are being reached or exceeded, and to evaluate whether staffing levels are sufficient to provide quality care. Utilisation, missed appointments, and service demand patterns will be comprehensively analysed to answer the questions.

Analysis

Three datasets were imported and examined using Pandas prior to analysis. The data underwent a thorough quality check, ensuring correct formats, columns, and the absence of missing or duplicate values. While no missing values were found, one dataset contained duplicates. Due to the absence of a unique identifier for records, it was deemed prudent to retain these duplicates to maintain data integrity. Descriptive statistics were subsequently generated.

Upon reviewing the metadata for all datasets, it became evident that data input is voluntary for practices, with no standardised requirements governing how practice staff record information. This variability raises concerns about data quality, leading to inconsistencies and potential reliability issues for analysis.

To ensure consistency, date formats across the datasets were standardised. Initially, the 'national categories' data set was analysed to calculate the distribution of total appointments between different categories in 8 different variables. User defined functions were written in Python, making use of Pandas, Seaborn and Matplotlib libraries to automatically calculate total appointments numbers and percentages in each category, and subsequently plot them on bar chart displaying the percentages for clarity.

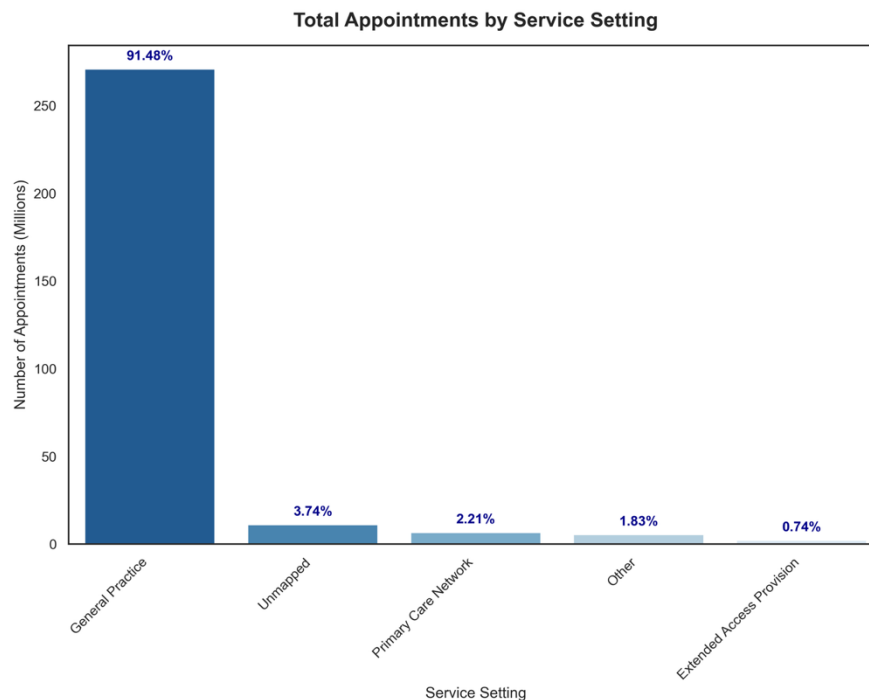
The dataset was subsequently filtered to analyse only data for North West London from January to June 2022, however this line of inquiry was deemed of little importance to the overall study, and was not pursued further.

Appointment trends were then analysed using the 'appointments regional' dataset, on both daily and monthly levels from August 2021 to June 2022, comparing them against the 1.2 million daily appointment capacity. User defined functions were again employed to plot temporal trends using Seaborn. Seasonal charts were annotated with a line illustrating NHS capacity for clear comparison. Appointments were also compared based on HCP (healthcare professional) type, appointment status, appointment mode and booking lead time, and their trends analysed temporally.

A closer analysis of appointment status was carried out by plotting stacked bar charts of how this variable is affected by booking lead time and appointment mode.

Additionally, a dataset of popular Twitter hashtags was imported and analysed to uncover any relevant patterns in discussions related to medical care.

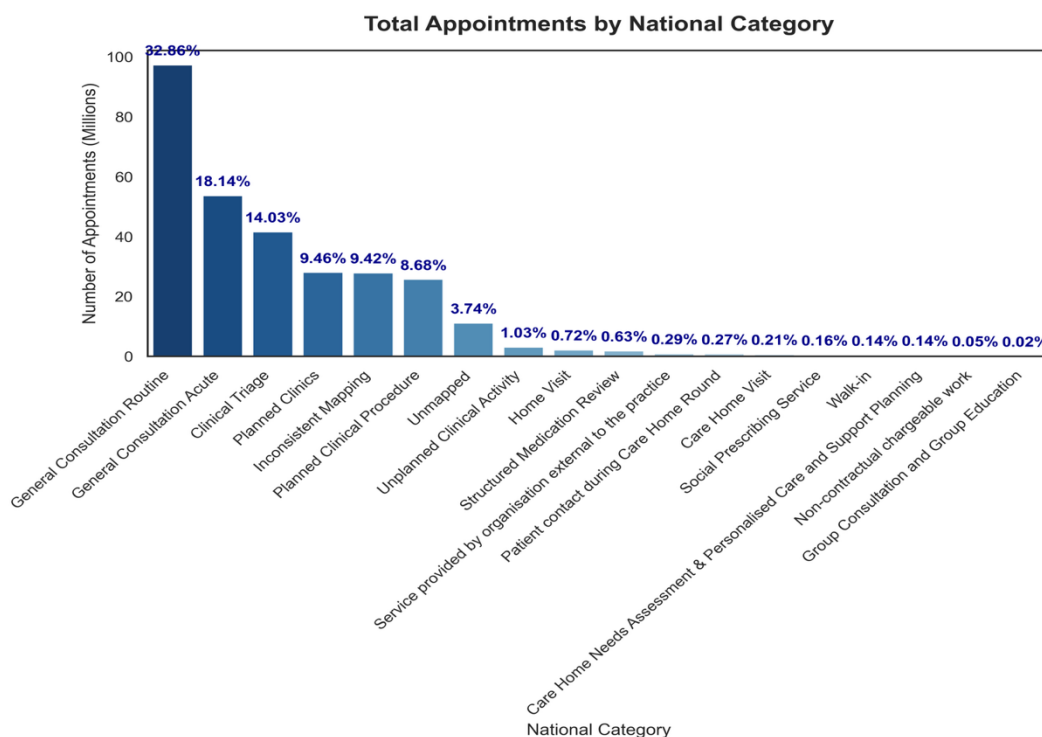
Results



Graph 1: bar chart showing total appointments by service setting

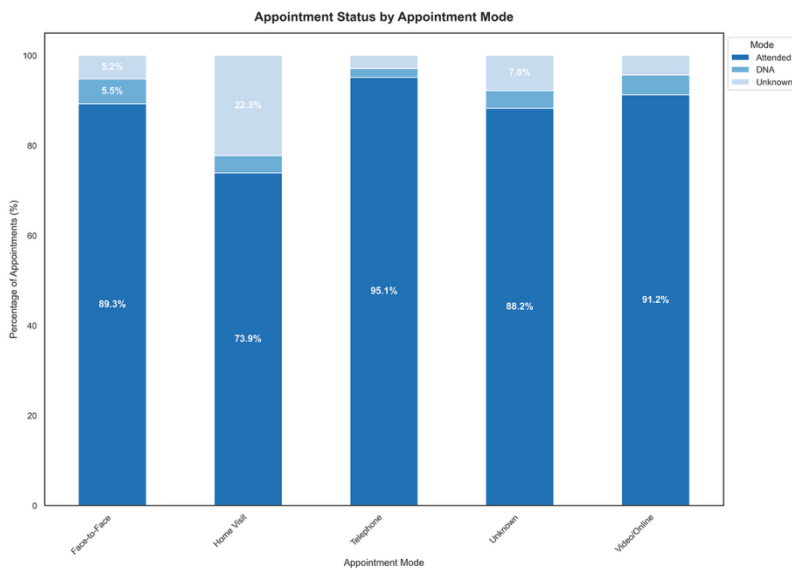
The initial analysis of appointments from August 2021 to June 2022 revealed that GPs account for over 91% of appointments across five service settings (Graph 1). In contrast, the Primary Care Network (PCN) and Extended Access Provision together contributed only 2.95%. PCNs aim to provide ‘proactive, personalised, coordinated, and integrated’ health and social care for patients (NHS).

Further breakdown of appointment types indicated that 65% consist of General Consultation Routine, Planned Clinics, and Clinical Triage. Unfortunately, 13% fell into categories of Inconsistent Mapping and Unmapped appointments, highlighting a critical data quality issue that must be addressed.

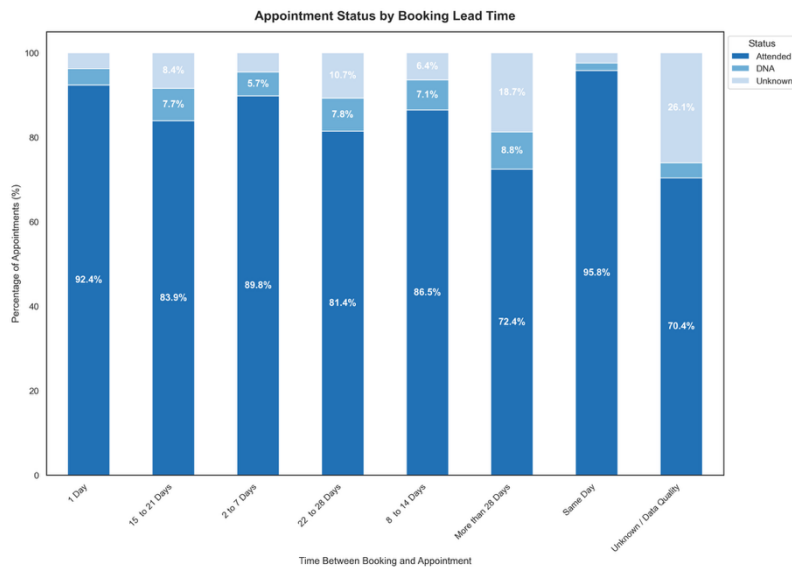


Graph 2: bar chart showing total appointments by national category

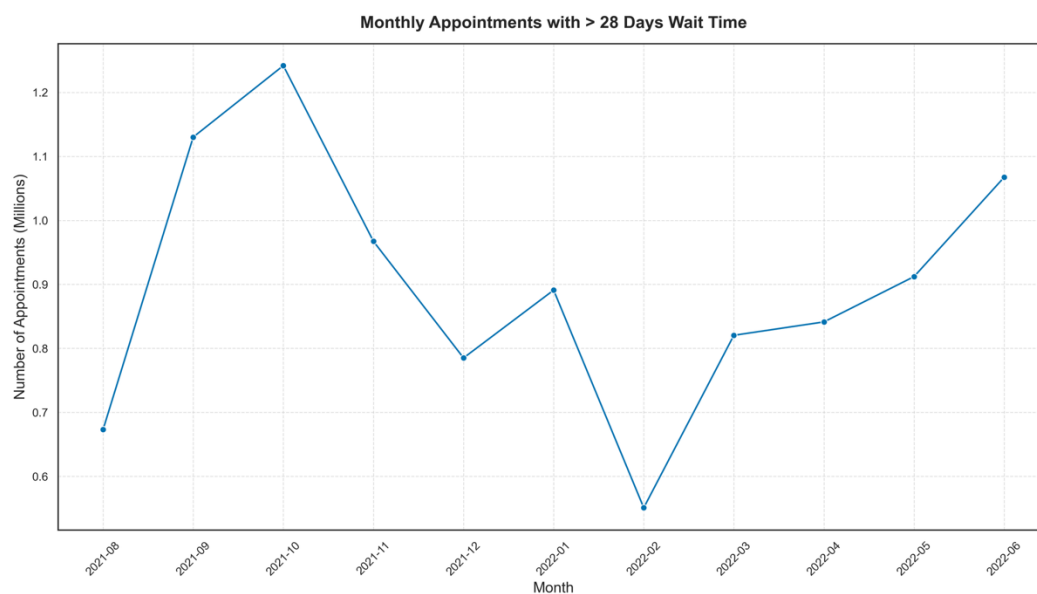
DNA Factor



↑ Graph 3: stacked bar chart showing appointment status by appointment mode



← Graph 4: stacked bar chart showing appointment status by booking lead time

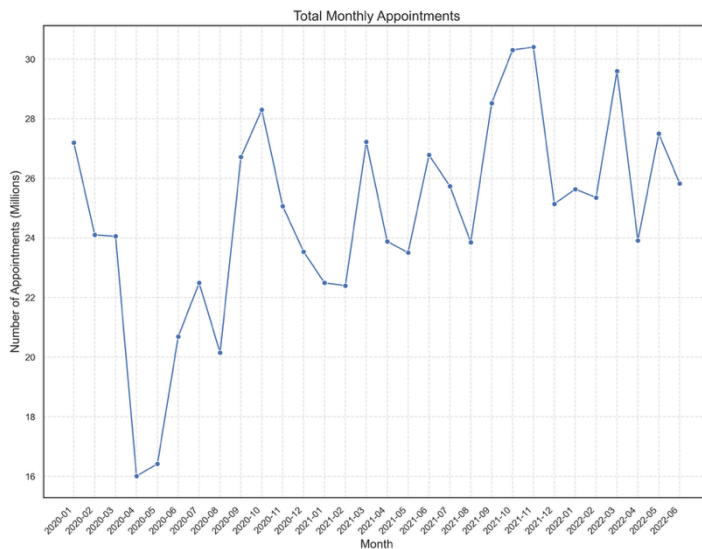


← Graph 5: line graph showing time series of monthly appointments with > 28 days waiting time

Despite over 90% of appointments being marked as attended, 4.1% are recorded as unattended. In October 2021, the busiest month analysed, this equated to approximately 1,565,624 missed appointments. With an average cost of £30 per appointment (NHS), the financial impact of missed appointments in October reached an astonishing £46,968,720. Notably, face-to-face appointments (Graph 3) and those with wait times exceeding 22 days (Graph 4) were the most frequently missed. Higher lead booking times coincide with periods of increased appointment volume, notably October 2021 (Graph 5).

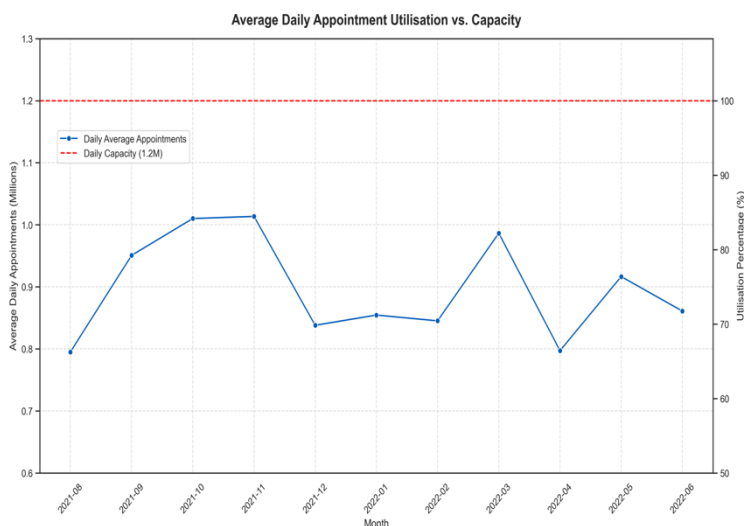
Therefore, it is clear that the busiest periods are also the ones which cost the NHS the most in missed appointments.

The 1.2M question



←Graph 6: line graph showing time series of total monthly appointments

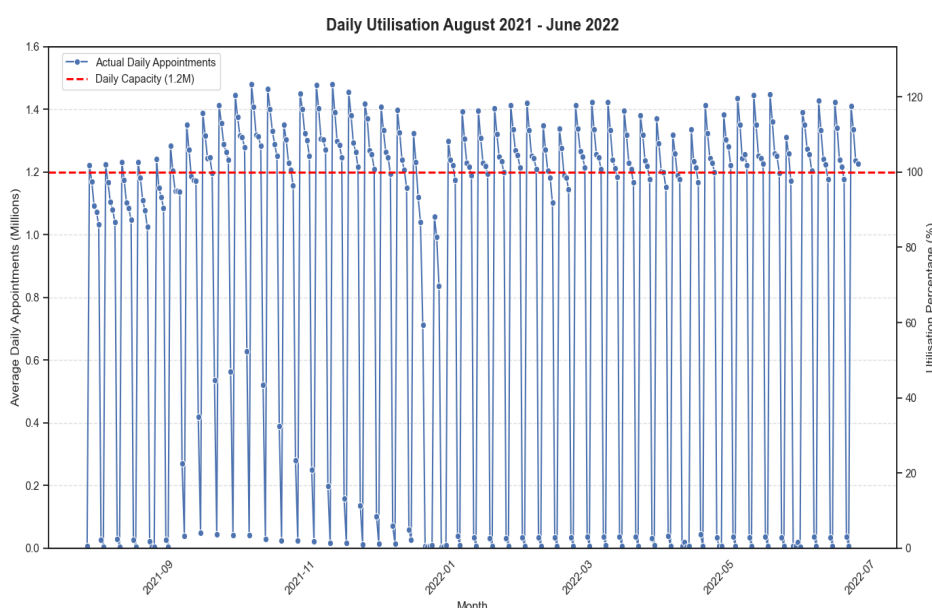
Graph 6 shows total monthly appointments from January 2020 – June 2022 and clearly illustrates changes in appointments corresponding to phases of the COVID-19 pandemic in the UK. Sharp decreases in appointments follow national lockdowns (e.g. March 2020 – first lockdown, December 2021 – announcement of Plan B measures), and increases in appointments coincide with the easing of lockdown restrictions.



Analysis of the data from August 2021 to June 2022 revealed that, on average, daily appointments do not exceed 85% of capacity.

←Graph 7: line graph showing time series of average daily appointment utilisation vs. capacity

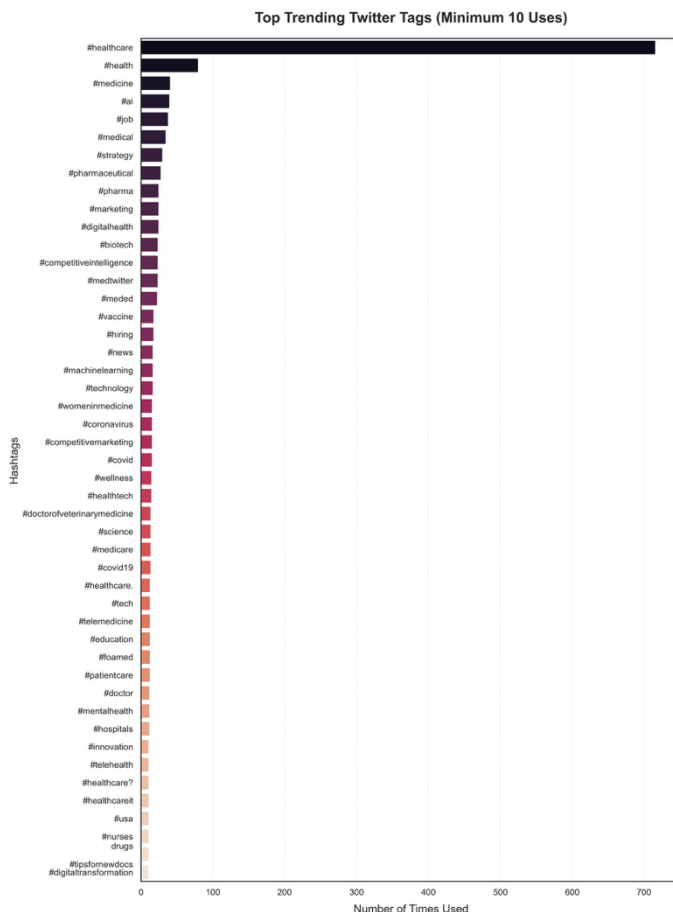
↓Graph 8: line graph showing time series of actual daily appointment utilisation vs. capacity



However, this average obscures true utilisation, as closer examination indicated that capacity is consistently exceeded, particularly on Mondays and Tuesdays in August 2021. For the remainder of the analysed timeframe, capacity was exceeded five days a week, with notable exceptions during the Christmas and Easter periods when appointment numbers dropped (Graph 8).

Twitter (now X)

Analysis of Twitter data indicated that 'healthcare' emerged as the most frequently used hashtag, though drawing meaningful conclusions from this trend proved challenging, beyond confirming that healthcare is a prominent topic of discussion. For further analysis, geotagging should be used to filter tweets by region, allowing social media pressure signals to be mapped against regional appointment utilisation data. A larger, more representative dataset is needed to provide deeper insight.



Graph 9: bar graph showing trending tags on Twitter

Recommendations

To enhance data reliability for future analyses, it is imperative that the NHS implement standardised data entry requirements across all practices, making data entry mandatory.

Furthermore, hiring additional staff is essential to alleviate the pressure on existing GPs. Specifically, the recruitment of weekend staff and the expansion of extended access provision appointments would help distribute utilisation more evenly throughout the week, preventing capacity overload, particularly at the start of the week. Enlisting weekend administrative staff to confirm attendance for appointments scheduled more than three weeks in advance would not only free up slots for urgent cases but also reduce missed appointments, potentially saving the NHS millions in costs associated with no-shows.

Increased utilisation of the Primary Care Network (PCN) for non-urgent cases is recommended to relieve pressure on GP clinics. Greater awareness of PCNs and how patients can access their services is essential.

Lastly, given the lower no-show rates associated with telephone appointments, it is advisable to expand the availability of these appointments for cases that do not necessitate in-person examinations. This is especially encouraged during periods of high appointment numbers. Increasing access to video/ online appointments is recommended for similar reasons, although not enough data on this mode is available as they only make up 0.49% of appointments.

Further investigation into actual staffing levels across different practice types and regional disparities is also warranted.

References

British Medical Association 2025, 'Safe working in general practice in England guidance', viewed 28 February 2026, <<https://www.bma.org.uk/advice-and-support/gp-practices/managing-workload/safe-working-in-general-practice>>

NHS England 2019, 'Missed GP appointments costing NHS millions', viewed 28 February 2026 <<https://www.england.nhs.uk/2019/01/missed-gp-appointments-costing-nhs-millions/>>

NHS England, 'Primary care networks', viewed 28 February 2026, <<https://www.england.nhs.uk/primary-care/primary-care-networks/>>

Appendix 1

5 Whys Framework

Problem statement:

The NHS does not have sufficient clarity on whether existing infrastructure and staffing capacity are adequate or whether expansion is required.

Why?

Because stakeholders disagree on whether current utilisation trends justify increasing resources, amid a declining GP workforce and a rising patient-to-GP ratio.

Why?

Because utilisation rates, missed appointments, and service demand patterns have not been comprehensively analysed.

Why ?

Because appointment data has not been systematically examined to determine whether capacity limits are being reached or exceeded.

Why?

Because there is no clear evaluation linking appointment demand, staffing levels, and capacity to determine whether current provision is sufficient.

Root Cause:

The NHS lacks a clear assessment of network utilisation trends and staffing sufficiency, making it difficult to determine whether existing capacity meets demand or whether additional resources are required.